

Question

A and **B** undergo [4+2] cycloaddition reaction to obtain compound **C** ($C_8H_{10}O_2$). **C** is hydrolyzed in acidic environment to **D** and methanol, and hydrolyzed in alkaline environment to **E** and methanol. **E** can spontaneously convert to **D** under acidic conditions. **C** reacts with excess methylmagnesium chloride in tetrahydrofuran to obtain **F**. **F** is treated with aqueous ammonium chloride solution to obtain the main product **G** ($C_9H_{14}O$) and a small amount of **H**. **G** can be converted relatively completely to **H** by treatment with concentrated acid, with the release of gas.

The following options are correct:

- A.** All other options are incorrect
- B.** **A**, **B**, **D** can all be deprotonated by a strong base (such as potassium tert-butoxide)
- C.** **H** does not possess strong toxicity.
- D.** Generation of **H** from **G** is accompanied by the formation of one molecule of a small molecule with a relative molecular mass of 44.
- E.** **F** possesses a chiral carbon atom.
- F.** **G**'s product in a strong alkaline solution (such as an ethanol solution of potassium hydroxide) has a tetrasubstituted double bond.
- G.** More than 3 correct items are in the options.
- H.** Exactly two of the other options are correct.

Answer

Correct Answer: **A**

Detailed Explanation

A and **B** undergo [4+2] cycloaddition reaction to obtain compound **C** ($C_8H_{10}O_2$), with an unsaturation degree of 4, and at least one six-membered ring, indicating that **C** has three unsaturated bonds;

CHECKPOINT

1 PTS

C has three unsaturated bonds and a six-membered ring

C is hydrolyzed in an acidic environment to **D** and methanol, and hydrolyzed in a basic environment to **E** and methanol. **E** can spontaneously convert to **D** in an acidic environment, which can be inferred that **C** contains a methyl ester substituent on the six-membered ring; then **D** is the corresponding carboxylic acid, and **E** is the sodium salt of the carboxylic acid.

CHECKPOINT

1 PTS

C contains a methyl ester substituent on the six-membered ring

CHECKPOINT

0.5 PTS

D is the corresponding carboxylic acid

CHECKPOINT

0.5 PTS

E is the sodium salt of the carboxylic acid

C reacts with excess methyl magnesium chloride in tetrahydrofuran to obtain **F**; it can be inferred that it is the reaction of the methyl ester group with the Grignard reagent, then the product **F** is the magnesium salt of a tertiary alcohol;

CHECKPOINT

1 PTS

F is the magnesium salt of a tertiary alcohol

F is treated with an aqueous solution of ammonium chloride to obtain the main product **G**, at which point the magnesium salt of the tertiary alcohol is hydrolyzed to obtain the tertiary alcohol. The molecular formula of **G** is ($C_9H_{14}O$), with an unsaturation degree of 3, thus containing a six-membered ring and two double bonds, and containing a tertiary alcohol structure transformed from a methyl ester substituent on a six-membered ring. Due to the characteristics of the cycloaddition reaction, the two double bonds generally constitute a 1,4-cyclohexadiene structure; thus, it is easy to draw the structure of **G** as CC(O)(C)C1=CCC=CC1.

CHECKPOINT

1 PTS

The two double bonds generally constitute a 1,4-cyclohexadiene structure due to the characteristics of the cycloaddition reaction

CHECKPOINT

1 PTS

The structure of **G** is CC(O)(C)C1=CCC=CC1

Deducing backwards, **F** is CC(C)(O[Mg]Cl)C1=CCC=CC1; **E** is O=C(O[Na])C1=CCC=CC1, **D** is O=C(O)C1=CCC=CC1,

CHECKPOINT

0.5 PTS

F is CC(C)(O[Mg]Cl)C1=CCC=CC1

CHECKPOINT

0.5 PTS

E is O=C(O[Na])C1=CCC=CC1

CHECKPOINT

0.5 PTS

D is O=C(O)C1=CCC=CC1

C is O=C(OC)C1=CCC=CC1, which exactly matches its molecular formula.

A and **B** undergo [4+2] to generate **C**, it is easy to obtain **A** and **B** as C=CC=C, C#CC(OC)=O, respectively.

CHECKPOINT

1 PTS

A is C=CC=C

CHECKPOINT

1 PTS

B is C#CC(OC)=O

A is butadiene, because the basicity required to break the carbon-hydrogen bond is much stronger than the basicity of potassium tert-butoxide, so it cannot be deprotonated by potassium tert-butoxide, so option B is incorrect.

CHECKPOINT

1 PTS

A is butadiene and cannot be deprotonated

G is treated with a strong acid, and the tertiary alcohol becomes a carbocation; at this time, there is no eliminable hydrogen on the ring, so only one molecule of propene can be eliminated to obtain a cyclohexadiene cation, which immediately aromatizes to generate benzene C1=CC=CC=C1; therefore, **H** is benzene C1=CC=CC=C1.

CHECKPOINT

1 PTS

H is benzene C1=CC=CC=C1

Benzene is toxic, option C is incorrect; the small molecule released from **G** to generate **H** is propene, with a relative molecular mass of 42, option D is incorrect.

CHECKPOINT

1 PTS

The small molecule released from **G** to generate **H** is propene

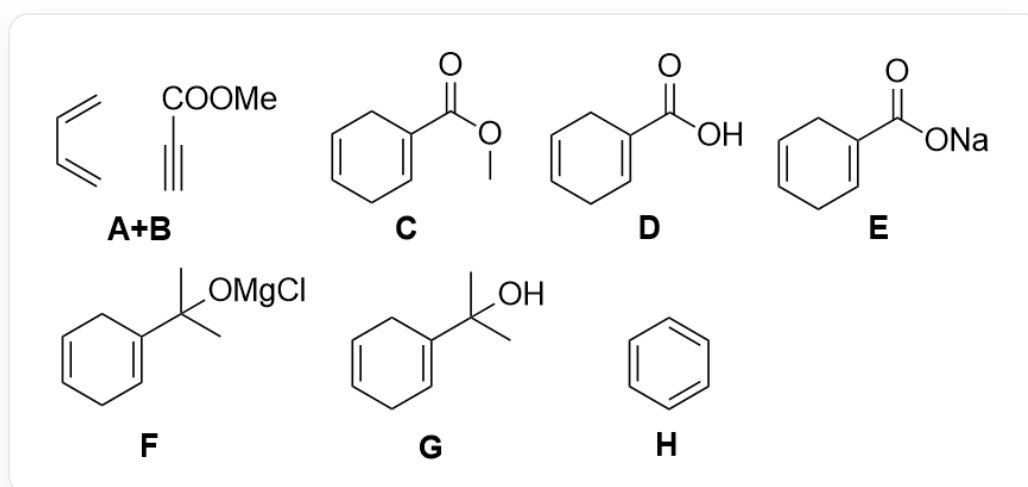
G is treated with a strong base, and the tertiary alcohol can only be eliminated to a terminal alkene, which is a disubstituted double bond, with the structure C=C(C)C1=CCC=CC1; this structure does not have a tetrasubstituted double bond, so option F is incorrect.

CHECKPOINT

1 PTS

The product of treating **G** with a strong base is C=C(C)C1=CCC=CC1

In summary, options B-F are all incorrect, and option A is correct.



This figure gives the structural formulas of the unknown structures **A** — **H** in this question, and their SMILES are all given in the analysis: **A** and **B** are C=CC=C and C#CC(OC)=O respectively, **C** is O=C(OC)C1=CCC=CC1, **F** is CC(C)(O[Mg]Cl)C1=CCC=CC1; **E** is O=C(O[Na])C1=CCC=CC1, **D** is O=C(O)C1=CCC=CC1, **G** is CC(O)(C)C1=CCC=CC1, **H** is C1=CC=CC=C1